

Normative basis. ACI 318-19 for required strength and detailing; ACI 421.1R-20 for general-shape perimeter mechanics and stud-rail guidance; ASTM A1044/A1044M-16a for headed shear stud assemblies. This document is version-locked. It is not an ACI 318-25 calculation.

Scope. Nonprestressed, solid two-way slabs without beams, reinforced with vertical smooth headed shear studs. Inch-pound equations only. Prestressed slabs, stirrups, shearheads, slabs with void systems, and seismic deformation checks require separate methods.

1 INPUT CONTRACT

The calculation shall not infer structural properties from display geometry. Every slab region shall reference one assigned slab type.

Input group	Required data
Slab type	h , f'_c , concrete density class or λ , top and bottom reinforcing bar size, spacing, layer order, cover, and f_y in both orthogonal directions.
Support	Actual loaded/reaction footprint; support centroid O ; interior, edge, or corner classification; slab boundaries; thickness changes; openings.
Actions	Signed V_u , $M_{sc,Ox}$, and $M_{sc,Oy}$ for every applicable factored load combination, all reported at the same point O and in the same axes.
Stud assembly	ASTM A1044/A1044M assembly, stud shaft diameter D , specified design yield strength f_{yt} , rail locations, stud coordinates, base-rail orientation, head/rail covers, and assembly height.

Validation. Require $h, d, f'_c, f_y, f_{yt} > 0$, $0.75 \leq \lambda \leq 1.0$, $0 < \lambda_s \leq 1.0$, consistent force/length units, and stud area consistent with the certified shaft diameter. Do not accept an independent diameter and area that disagree.

2 EFFECTIVE DEPTH

For each load combination, identify the actual tension reinforcement relative to the extreme compression face. Using the reinforcement centroids,

$$d = \frac{d_x + d_y}{2}. \quad (1)$$

d_x and d_y are distances from the extreme compression face to the centroids of the tension reinforcement in the two slab directions. Layer order matters. For moment reversal, the bottom mat can govern and d shall be recomputed. Bar spacing does not directly enter ACI 318-19 punching strength, but it does enter the companion flexural-reinforcement checks in Section 10.

3 CRITICAL SECTIONS

3.1 INNER SECTIONS

For two-way shear, select the polygon having minimum effective perimeter b_o that is not closer than $d/2$ to:

- edges or corners of the column, concentrated load, or reaction area; and
- changes in slab thickness, including drop panels, capitals, or shear caps.

Check every applicable section. A circular or regular polygonal support may be replaced by a square of equal area when permitted by ACI 318-19 22.6.4.1.2.

At a slab edge, the critical section may terminate at the free edge. Do not count a fictitious segment along the free edge.

3.2 OPENINGS

If the nearest opening boundary is closer than $4h$ to the periphery of the column, load, or reaction area, draw tangents from the support/load centroid to the opening. The portion of each critical perimeter between those tangents is ineffective. Apply the rule independently to the inner and outer sections. Openings at $4h$ or farther do not reduce b_o under ACI 318-19 22.6.4.3.

3.3 OUTER SECTION WITH STUDS

Also check a polygon at $d/2$ beyond the actual outermost peripheral line of studs, selected to minimize b_o . Build this perimeter from the real endpoint of every rail/face. A single maximum stud radius is not a valid substitute for the actual layout. At free edges, terminate the effective perimeter at the slab edge.

4 NET ACTIONS AT EACH PERIMETER

For every load combination and every critical perimeter P , use the net factored actions crossing that perimeter:

$$V_u(P) = R_u - \int_{A_{\text{inside}}(P)} q_u dA. \quad (2)$$

Apply the same equilibrium to both moment components. If the loads inside a perimeter are intentionally ignored, report that assumption; do not silently reuse an inner demand at the outer perimeter.

Translate moments from support centroid O to critical-section centroid C . Using the ACI 421 sign convention,

$$M_{sc,\bar{x}} = M_{sc,O\bar{x}} + V_u \bar{y}_O, \quad M_{sc,\bar{y}} = M_{sc,O\bar{y}} + V_u \bar{x}_O, \quad (3)$$

where $(\bar{x}_O, \bar{y}_O)$ locates O from C . Then rotate both geometry and moments to the centroidal principal axes:

$$\begin{aligned} x &= \bar{x} \cos \theta + \bar{y} \sin \theta, & y &= -\bar{x} \sin \theta + \bar{y} \cos \theta, \\ M_{sc,x} &= M_{sc,\bar{x}} \cos \theta - M_{sc,\bar{y}} \sin \theta, & M_{sc,y} &= M_{sc,\bar{x}} \sin \theta + M_{sc,\bar{y}} \cos \theta. \end{aligned} \quad (4)$$

5 GENERAL-SHAPE SECTION PROPERTIES

Partition the effective perimeter into straight segments with endpoints (x_i, y_i) and (x_j, y_j) and length L_k . First calculate the line centroid:

$$b_o = \sum_k L_k, \quad A_c = b_o d, \quad \bar{x} = \frac{\sum_k L_k (x_i + x_j)/2}{b_o}, \quad \bar{y} = \frac{\sum_k L_k (y_i + y_j)/2}{b_o}. \quad (5)$$

Shift all endpoint coordinates to $(\bar{x}, \bar{y}) = (0, 0)$. In any centroidal axes,

$$J_{\bar{x}} = d \sum_k \frac{L_k}{3} (y_i^2 + y_i y_j + y_j^2), \quad (6)$$

$$J_{\bar{y}} = d \sum_k \frac{L_k}{3} (x_i^2 + x_i x_j + x_j^2), \quad (7)$$

$$J_{\bar{x}\bar{y}} = d \sum_k \frac{L_k}{6} (2x_i y_i + x_i y_j + x_j y_i + 2x_j y_j). \quad (8)$$

Rotate to principal axes so $J_{xy} = 0$. A nonzero J_{xy} is not a warning condition: it requires the inertia tensor and both moment components to be transformed. The principal-axis values J_x and J_y shall be positive.

6 FACTORED SHEAR-STRESS DISTRIBUTION

In centroidal principal axes, with signs defined by equilibrium,

$$v_u(x, y) = \frac{V_u}{b_o d} + \frac{\gamma_{vx} M_{sc,xy}}{J_x} + \frac{\gamma_{vy} M_{sc,yx}}{J_y}. \quad (9)$$

Evaluate the signed expression at every endpoint of every effective segment for every factored load combination:

$$v_{u,\max} = \max_{\text{combos, endpoints}} |v_u(x, y)|. \quad (10)$$

Because the assumed stress is linear on each straight segment, its extrema occur at segment endpoints. The calculator shall use the conservative endpoint value; it shall not apply the optional isolated-peak redistribution discussed in ACI 421 unless that option is separately implemented, limited, and reported.

7 MOMENT TRANSFER FRACTIONS

Let l_x and l_y be the full projections of the current critical perimeter onto its principal x and y axes. Recompute these values and the coefficients for the outer section.

7.1 INTERIOR CONNECTION

$$\gamma_{vx} = 1 - \frac{1}{1 + \frac{2}{3}\sqrt{l_y/l_x}}, \quad \gamma_{vy} = 1 - \frac{1}{1 + \frac{2}{3}\sqrt{l_x/l_y}}. \quad (11)$$

7.2 EDGE CONNECTION

For a standard three-sided perimeter, take x normal to the free edge and y parallel to it:

$$\gamma_{vx} = 1 - \frac{1}{1 + \frac{2}{3}\sqrt{l_y/l_x}}, \quad (12)$$

$$\gamma_{vy} = \begin{cases} 0, & l_x/l_y \leq 0.2, \\ 1 - \frac{1}{1 + \frac{2}{3}\sqrt{l_x/l_y - 0.2}}, & l_x/l_y > 0.2. \end{cases} \quad (13)$$

7.3 CORNER CONNECTION

For the standard two-sided perimeter, label the principal axes per ACI 421 with $l_x \leq l_y$:

$$\gamma_{vx} = 0.40, \quad \gamma_{vy} = \begin{cases} 0, & l_x/l_y \leq 0.2, \\ 1 - \frac{1}{1 + \frac{2}{3}\sqrt{l_x/l_y - 0.2}}, & l_x/l_y > 0.2. \end{cases} \quad (14)$$

Do not classify a perimeter from a bounding box. If openings, multiple free edges, or a nonstandard footprint create a topology that does not match the ACI 421 interior/edge/corner cases, require an engineer-specified, documented method instead of inventing γ_v .

The basic ACI 318 moment split is $\gamma_v = 1 - \gamma_f$. Do not reduce γ_v under the optional ACI 318-19 8.4.2.2.4 redistribution unless its v_{uv} , strain, and flexural-strength conditions are all checked and $\gamma_f M_{sc}$ is transferred to the slab reinforcement.

8 CONCRETE STRENGTH WITHOUT STUDS

ACI 318-19 22.6.3 limits the concrete strength used to calculate v_c :

$$f'_{c,v} = \min(f'_c, 10,000 \text{ psi}). \quad (15)$$

Use the specified f'_c for checks that are not calculations of v_c , unless their governing provision states otherwise. For nonprestressed members, in psi,

$$v_c = \min \left\{ \begin{aligned} &4\lambda_s \lambda \sqrt{f'_{c,v}} \\ &2 \left(1 + \frac{2}{\beta} \right) \lambda_s \lambda \sqrt{f'_{c,v}} \\ &\left(2 + \frac{\alpha_s d}{b_o} \right) \lambda_s \lambda \sqrt{f'_{c,v}} \end{aligned} \right\}, \quad (16)$$

$$\lambda_s = \min \left(1.0, \sqrt{\frac{2}{1 + d/10}} \right). \quad (17)$$

β is the long-to-short dimension ratio of the effective column, loaded, or reaction area. For a nonrectangular area, use the ACI effective loaded-area definition, not the critical-perimeter bounding box. Use $\alpha_s = 40, 30, 20$ for interior, edge, and corner connections.

$$\boxed{v_{u,\max} \leq \phi v_c}, \quad \phi = 0.75. \quad (18)$$

9 HEADED SHEAR STUD STRENGTH

9.1 MATERIAL AND AREA

Stud assemblies shall conform to ASTM A1044/A1044M. Use the specified design yield strength certified for the assembly and accepted by the project design basis:

$$f_{yt,v} = \min(f_{yt,\text{specified}}, f_{yt,\text{qualified}}, f_{yt,\text{project limit}}, 60,000 \text{ psi}). \quad (19)$$

The 60 ksi maximum follows ACI 318-19 22.6.3.2 and 20.2.2.4. Do not use a higher value in v_s . Where the product basis specifies 51 ksi, use 51,000 psi.

$$A_{\text{stud}} = \frac{\pi D^2}{4}, \quad A_v = N_p A_{\text{stud}}, \quad (20)$$

where N_p is the number of studs crossing one geometrically similar peripheral line. A_v is not the area of every stud on every rail.

9.2 INNER STUD-REINFORCED SECTION

$$v_{c,\text{stud}} = \min \left\{ \begin{array}{l} 3\lambda_s \lambda \sqrt{f'_{c,v}} \\ 2 \left(1 + \frac{2}{\beta} \right) \lambda_s \lambda \sqrt{f'_{c,v}} \\ \left(2 + \frac{\alpha_s d}{b_o} \right) \lambda_s \lambda \sqrt{f'_{c,v}} \end{array} \right\}, \quad v_s = \frac{A_v f_{yt,v}}{b_o s}. \quad (21)$$

Once studs are required, the design ratio shall satisfy both strength and the minimum reinforcement:

$$\left(\frac{A_v}{s} \right)_{\text{req}} = \max \left[\frac{b_o}{f_{yt,v}} \max \left(\frac{v_{u,\text{max}}}{\phi} - v_{c,\text{stud}}, 0 \right), \frac{2\sqrt{f'_c} b_o}{f_{yt,v}} \right]. \quad (22)$$

The provided ratio is valid only for a layout with constant radial spacing s and a defined A_v on each peripheral line. Variable rail spacings or missing studs break that idealization and shall be rejected or checked line-by-line with a separately validated method.

Both inner limits are mandatory:

$$\boxed{v_{u,\text{max}} \leq \phi(v_{c,\text{stud}} + v_s)}, \quad \boxed{v_{u,\text{max}} \leq 8\phi\sqrt{f'_c}}. \quad (23)$$

9.3 OUTER SECTION

Stud contribution is zero outside the reinforced zone:

$$v_{c,\text{out}} = 2\lambda_s \lambda \sqrt{f'_{c,v}}, \quad \boxed{v_{u,\text{max,out}} \leq \phi v_{c,\text{out}}}. \quad (24)$$

Use the outer section's own b_o , centroid, principal axes, J_x , J_y , l_x , l_y , γ_{vx} , γ_{vy} , net force, and net moments.

9.4 SIZE-EFFECT EXCEPTION

The default is the calculated λ_s . It may be set to 1.0 only when smooth headed shear studs are designed and detailed per ACI 318-19 8.7.7, the smooth shaft length of every applicable stud is at most 10 in., and

$$\frac{A_v}{s} \geq \frac{2\sqrt{f'_c} b_o}{f_{yt,v}}. \quad (25)$$

Apply the accepted value consistently to the inner and outer concrete terms. Overall assembly height and smooth shaft length are different dimensions.

10 STUD DETAILING RULES

Rule	Acceptance criterion
Product	ASTM A1044/A1044M assembly; studs perpendicular to slab plane.
First peripheral line	Face-to-stud distance measured normal to support face, $s_o \leq d/2$.
Radial spacing, lower demand	Constant $s \leq 3d/4$ when $v_{u,\max} \leq \phi 6\sqrt{f'_c}$.
Radial spacing, higher demand	Constant $s \leq d/2$ when $v_{u,\max} > \phi 6\sqrt{f'_c}$.
Transverse spacing	Adjacent studs on the first peripheral line: $g \leq 2d$, measured parallel to the relevant column face. Check each face; do not connect points across free edges or openings.
Extent	Every face/rail shall reach the actual outer stud line used to construct the passing outer critical section.
Assembly height	$h_{\text{assembly}} \geq h - c_{\text{top}} - c_{\text{rail}} - d_b/2$.
Peripheral lines	At least two peripheral lines when studs are provided (GUIDE : ACI 421.1R layout recommendation).

Check dimensions against actual stud coordinates and actual slab boundaries. Do not infer compliance from rail count, maximum radius, or a rendered symbol.

11 COMPANION FLEXURAL CHECKS

Studs do not eliminate slab-column flexural-transfer requirements.

11.1 MOMENT TRANSFERRED BY FLEXURE

In each direction, reinforcement within the effective slab width b_{slab} shall satisfy

$$\phi_f M_n \geq \gamma_f M_{sc}. \quad (26)$$

b_{slab} is the support width plus the permitted distance on each side under ACI 318-19 8.4.2.2.3, generally limited to $1.5h$ on each side and further limited by slab edges or drop/cap geometry.

11.2 FLEXURE-DRIVEN PUNCHING MINIMUM

For a nonprestressed two-way slab, calculate the direct shear stress without moment transfer:

$$v_{uv} = \frac{|V_u|}{b_o d}. \quad (27)$$

If

$$v_{uv} > \phi 2\lambda_s \lambda \sqrt{f'_c}, \quad (28)$$

provide bonded flexural tension reinforcement over b_{slab} :

$$A_{s,\text{min,FDPS}} = \frac{5v_{uv} b_{\text{slab}} b_o}{\phi \alpha_s f_y}. \quad (29)$$

Check both slab directions, the actual tension face for each load combination, bar development, and the ordinary slab minimum reinforcement. For uniform bars, $A_{s,\text{prov}} \approx A_{\text{bar}} b_{\text{slab}}/s_{\text{bar}}$; use actual bar coordinates when available.

12 DETERMINISTIC CHECK SEQUENCE

- 1 Validate assigned slab type, materials, units, reinforcement layers, support footprint, boundaries, openings, actions, and stud certification.
- 2 For each load combination, determine the tension face and d_x, d_y, d .
- 3 Construct every inner critical section; apply free-edge and $4h$ opening rules; calculate net actions.
- 4 Calculate centroid and full inertia tensor; rotate geometry and actions to principal axes; compute γ_v from the actual perimeter topology.
- 5 Evaluate signed stress at every segment endpoint; retain the largest absolute value and its load combination, perimeter, segment, and coordinate.
- 6 Check concrete-only strength. If it fails, require the stud compression cap before designing A_v/s .
- 7 Check inner stud strength, minimum A_v/s , product properties, height, s_o, s, g , and at least two peripheral lines.
- 8 Construct the minimum outer perimeter from real outermost studs; recompute all geometry, actions, coefficients, and stress; check concrete-only strength.
- 9 Check $\gamma_f M_{sc}$ flexural transfer and ACI 318-19 8.6.1.2 flexure-driven punching reinforcement.
- 10 Pass only when every factored combination and every required inner and outer section passes. Report all governing values and code references.

13 REQUIRED RESULT RECORD

The calculation output shall retain, not merely display:

- code edition; slab type ID; $h, f'_c, \lambda, d_x, d_y, d$;
- effective perimeter segments before and after edge/opening reductions;
- $b_o, A_c, C, J_{\bar{x}}, J_{\bar{y}}, J_{\bar{x}\bar{y}}, \theta, J_x, J_y, l_x, l_y$;
- net $V_u, M_{sc,x}, M_{sc,y}$ and γ_{vx}, γ_{vy} by combination;
- controlling endpoint and signed stress components;
- each concrete-strength branch, $f'_{c,v}, \lambda_s, \beta$, and α_s ;
- $D, A_{stud}, N_p, A_v, s, A_v/s, f_{yt}, f_{yt,v}$ and every detailing limit;
- inner strength, compression cap, outer strength, flexural transfer, and flexure-driven punching checks.

14 CODE MAP

Subject	Reference
Moment transfer and shear stress	ACI 318-19 8.4.2.2 and 8.4.4.2; ACI 421.1R-20 Chs. 4 and 7, App. B
Minimum slab flexural steel	ACI 318-19 8.6.1.2
Stud placement and assembly height	ACI 318-19 8.7.7.1
Critical sections and openings	ACI 318-19 22.6.4
Concrete strength without studs	ACI 318-19 22.6.5
Strength with headed studs	ACI 318-19 22.6.6 and 22.6.8
Material-strength limits	ACI 318-19 22.6.3 and 20.2.2.4
Strength reduction factor	ACI 318-19 Table 21.2.1
Stud product requirements	ACI 318-19 20.4; ASTM A1044/A1044M-16a